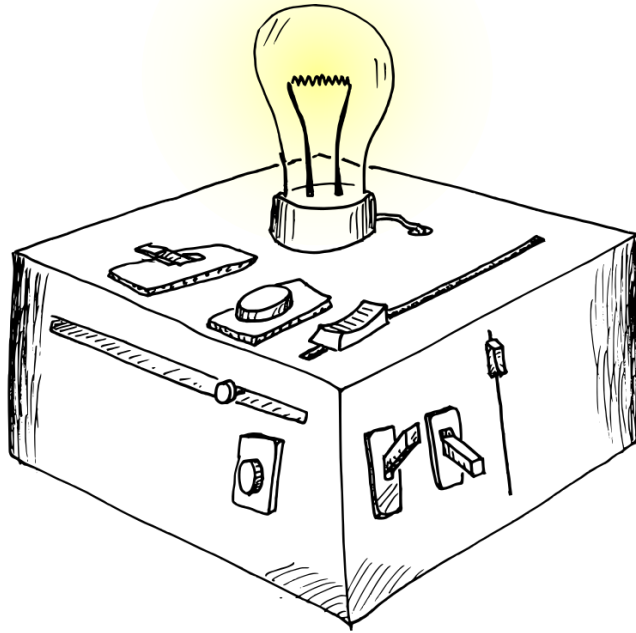


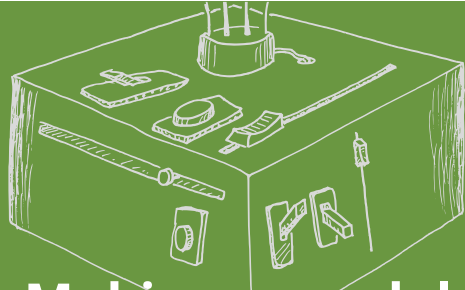
Model to meaning

How to interpret generalized multilevel models in 

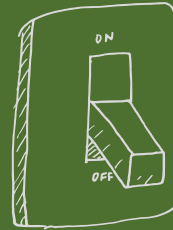


Luke Watson

Outline



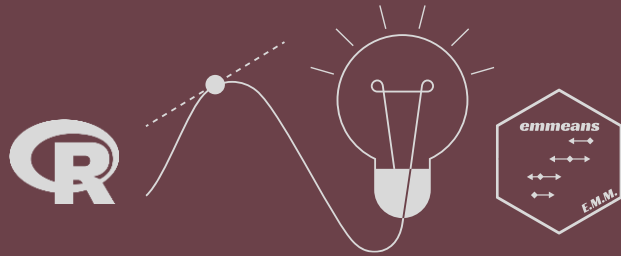
Making a model



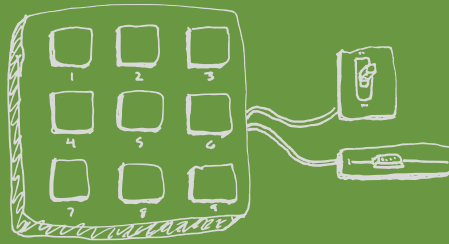
Categorical contrasts



Marginal effects



Implementation



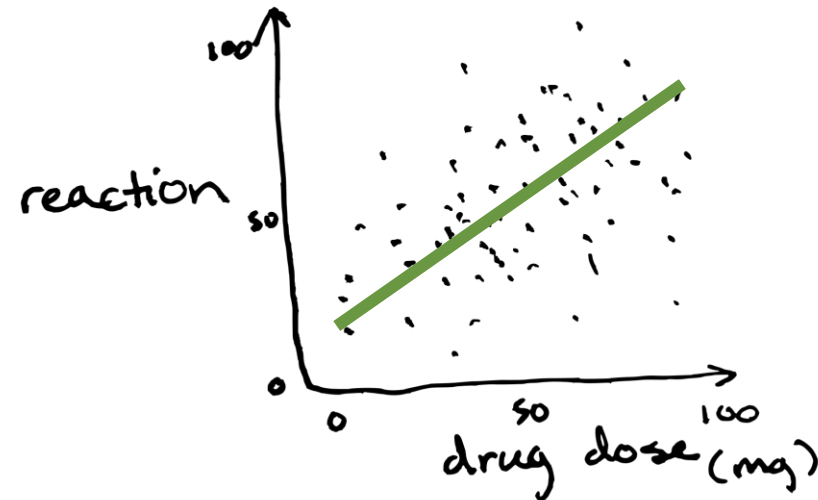
Extras

Events and encounters

- › As researchers, we create models to make sense of how events and encounters are related to each other
- › Sometimes it is simple, we record information about how individuals respond to encountering the **presence or absence** of a single event or **gradient** of that event
 - › Like gradient of drug doses, presence of CS+ or absence of CS+, etc.

Simple model

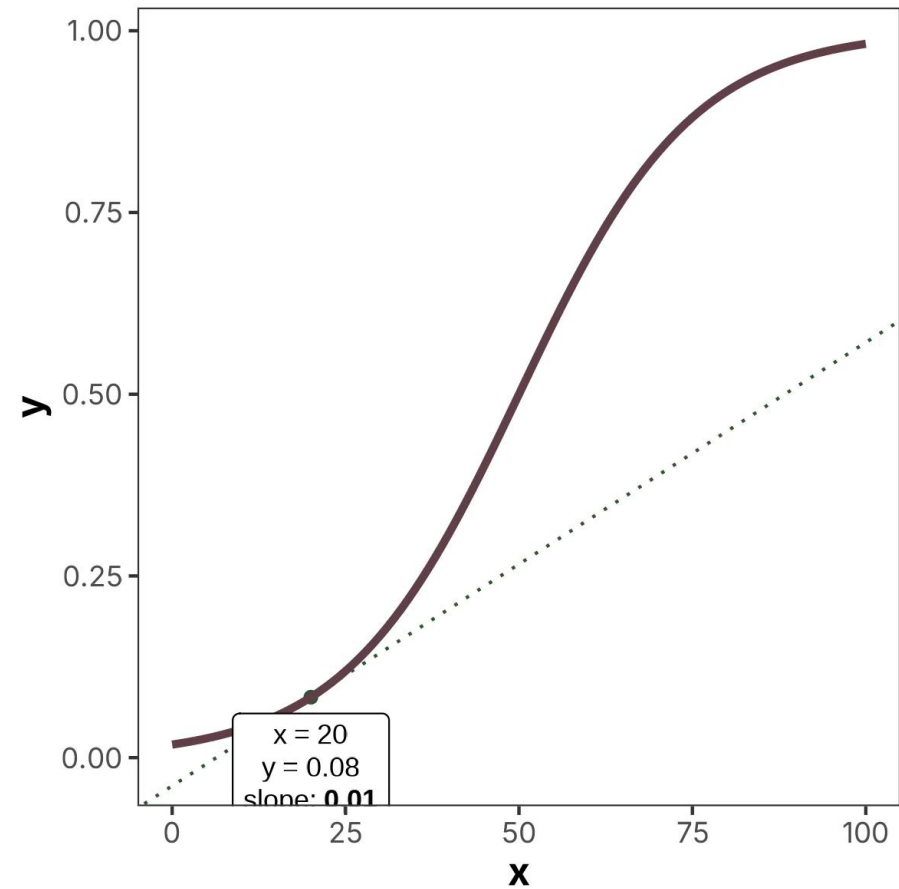
$$\text{reaction}_i = \beta_{\text{drug}_i} \times \text{drug}_i$$



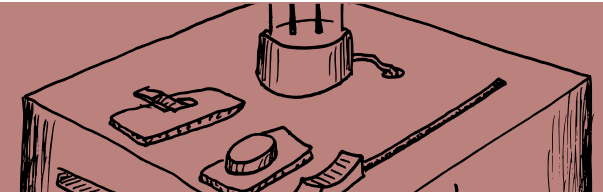
Effect of drug dose on reaction remains constant across doses,
equal to β (slope)

More complexity

- › A harder scenario is a non-linear model, where the effect of x on y changes across values of x
- › A single β slope no longer tells us exactly what is happening

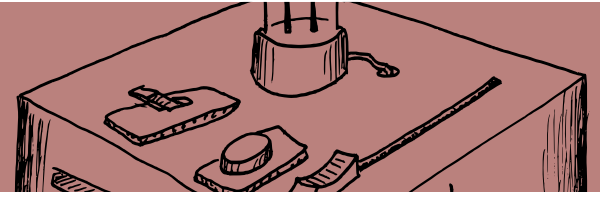


Research models

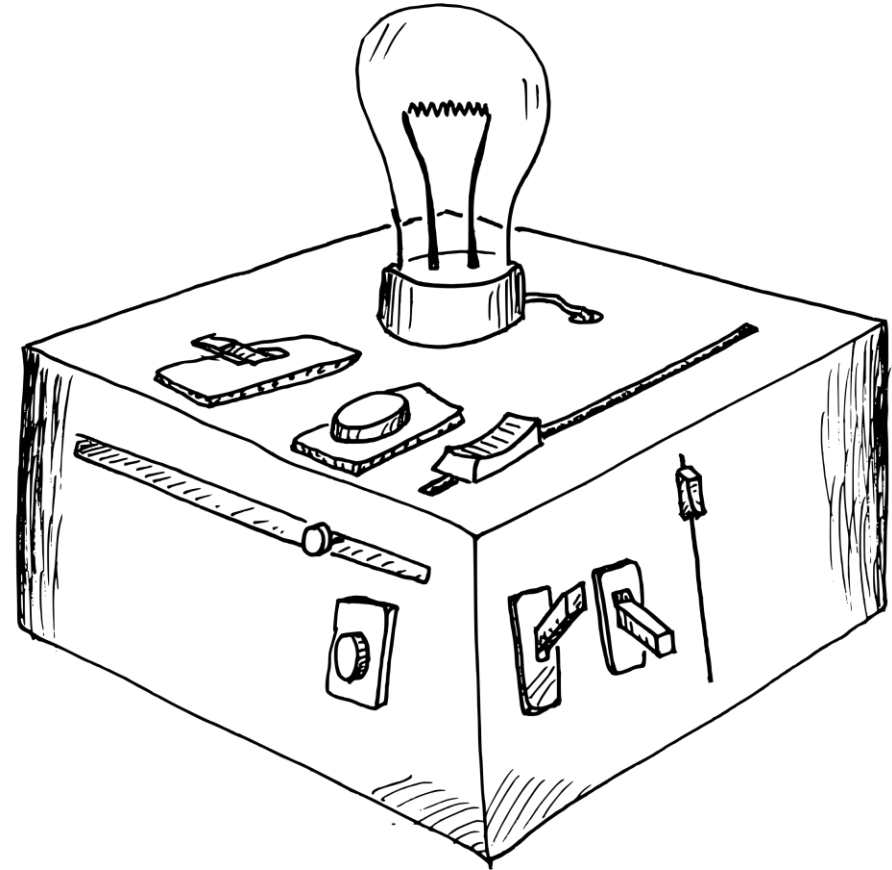


- › The models we use are often
 - › **Generalized** LMs (model non-continuous/normal outcomes)
 - › Contain several predictors and their interactions, both categorical and continuous
 - › Contain grouped observations (repeated measures of the same subjects/stimuli)
- › $Reaction\ time_i \sim Gamma(\mu_i, shape)$
 - › $\mu_i = \beta_0 + \beta_{0_{subject}} + (\beta_{dose} + \beta_{dose_{subject}}) * dose_i + (\beta_{condition} + \beta_{condition_{subject}}) * condition_i + (\beta_{dose*condition} + \beta_{dose*condition_{subject}}) * dose_i * condition_i$

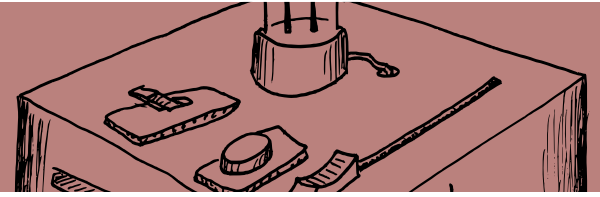
The typical model



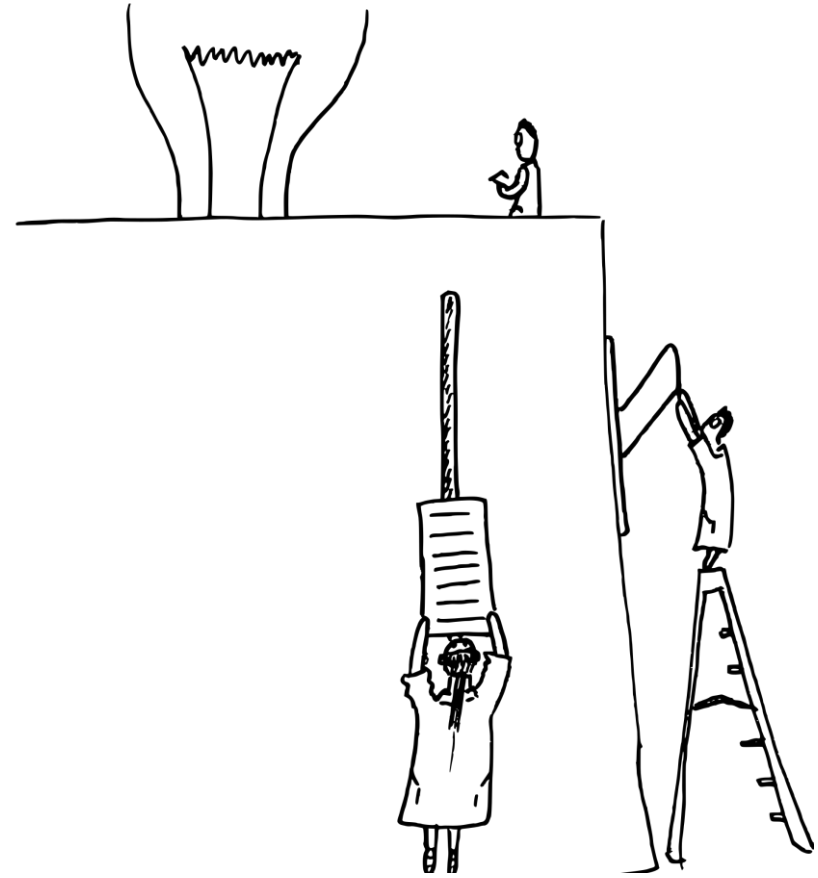
- › Models like ours can be thought of like little machines we create
- › We assemble **sliders** and **switches** (predictors) connected to an **output device** (outcome)
- › Now, how do these components affect our output?



Investigating the model



- › We can be structured about our investigation
 - › Observe the effect of moving one slider while holding all other switches and sliders constant
- › This is easier to intuit than a table of numbers vaguely related to how our machine functions

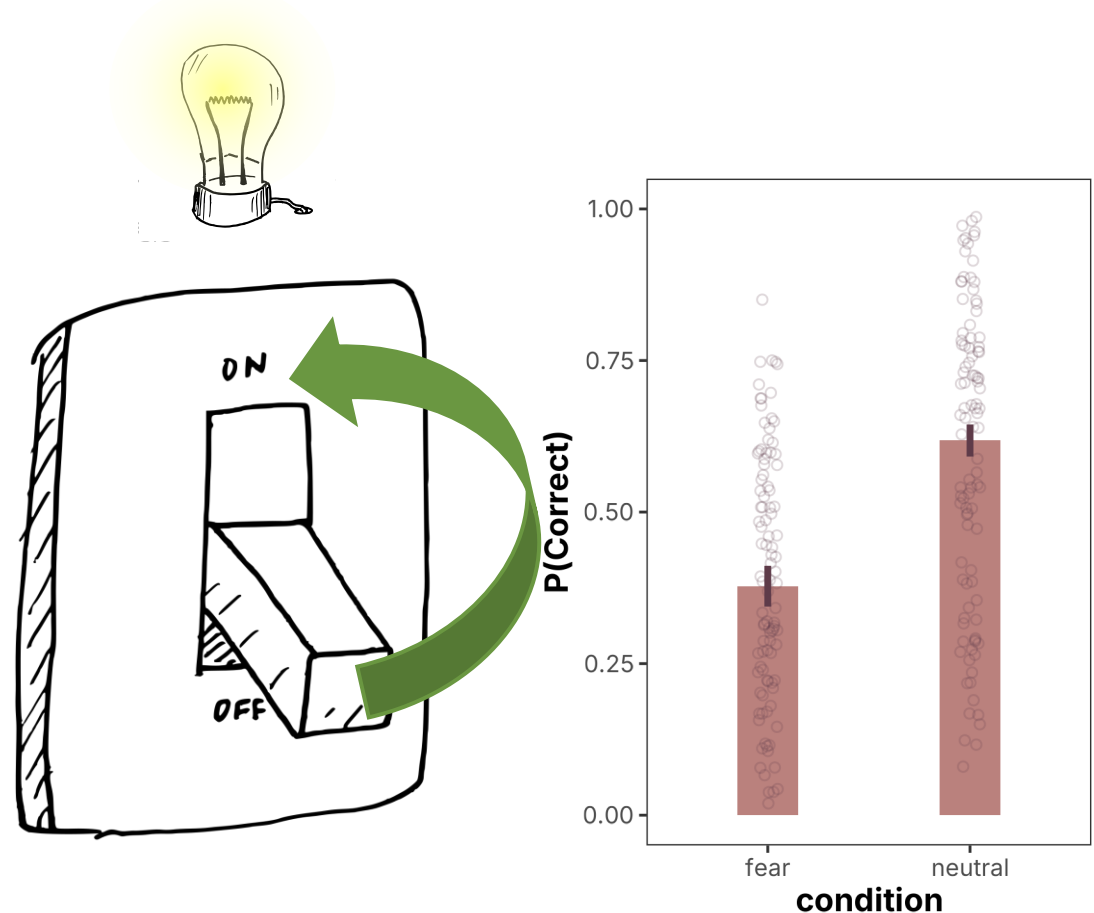


Marginal effects

- › In the stats world, this is known as a marginal effect
- › The **individual effect** of a predictor on the outcome while **holding all other variables constant**
- › There are lots of different ways to accomplish this and plenty of names they go by (just like modeling itself)!

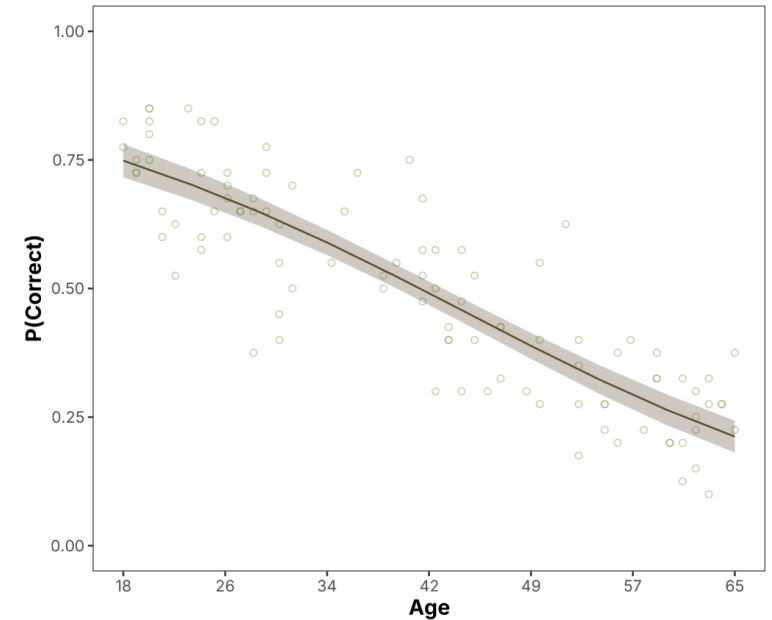
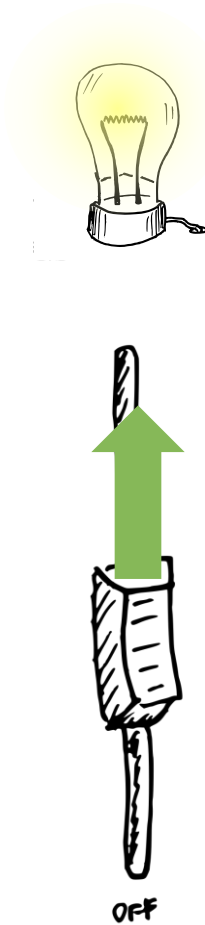
Categorical contrasts

- › The effect of turning a predictor from **off** to **on**
 - › These are our switches
- › Change in reaction time with and without a drug
- › Usually visualized as a bar plot



True marginal effects

- › A true marginal effects concern how **small changes** in our predictor affects our outcome
 - › Aka our sliders
- › Change in reaction time as dose increases gradually
- › Usually visualized with a line plot

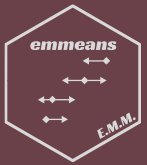


Implementation

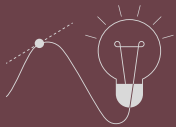
- › We can compute categorical contrasts and marginal effects with our models in R in a couple of ways



Base R: `fitted()` and `predict()` with manual post-processing

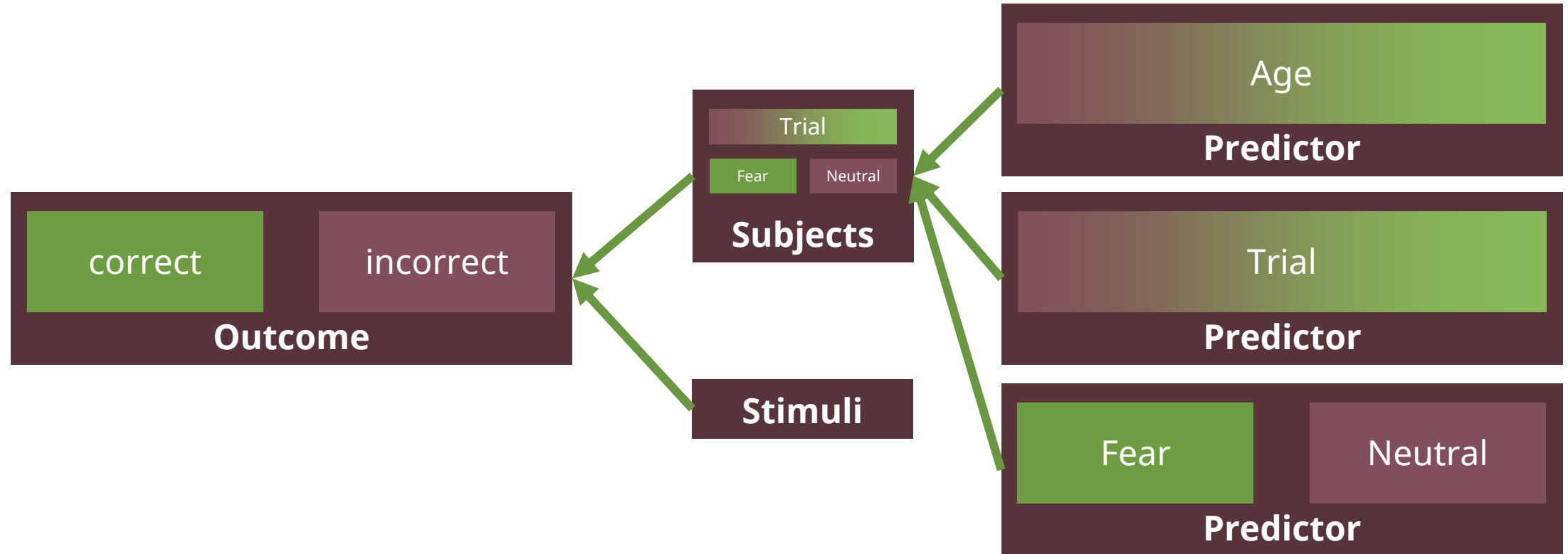


emmeans: library designed for performing statistical contrasts and marginal effects



marginaleffects: library works like emmeans but also optimized for visualization

Example model



> **glmer**

- > formula = Correct ~ Age * Trial * Condition + (1 | Stimulus) + (Trial * Condition | Subject)
- > family = 'binomial'

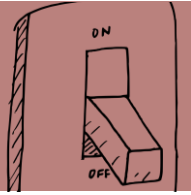
Example model results

> glmer

- > formula = Correct ~ Age * Trial * Condition + (1 | Stimulus) + (Trial * Condition | Subject)
- > family = 'binomial'

term	b	SE	t	p
<i>(Intercept)</i>	-0.01	0.05	-0.18	0.859
<i>age</i>	-0.85	0.05	-16.10	<.001
<i>trial</i>	0.10	0.05	2.30	0.021
<i>condition</i>	-0.49	0.04	-11.90	<.001
<i>age x trial</i>	0.02	0.05	0.44	0.658
<i>age x condition</i>	0.19	0.04	4.47	<.001
<i>trial x condition</i>	0.00	0.04	0.02	0.984
<i>age x trial x condition</i>	-0.11	0.04	-2.71	0.007

Contrasts



› Base R predict/fitted

› code

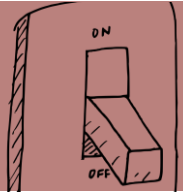
```
data_with_predictions <- my_data >
  add_column(predicted = fitted(glmer_model))

data_with_predictions >
  group_by(condition) >
  summarize(mean_p_correct = mean(predicted))
```

› output

```
> condition  mean_p_correct
1 fear              0.393
2 neutral              0.595
```

Contrasts



> emmeans emmeans()

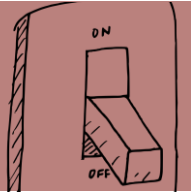
> code

```
model_predictions <- emmeans(  
  glmer_model,  
  ~ condition * age * trial,  
  regrid = 'response'  
)
```

> output

condition	x_age	x_trial	prob	SE	df	asympt.LCL	asympt.UCL
fear	-2.02e-16	0	0.378	0.0172	Inf	0.344	0.411
neutral	-2.02e-16	0	0.618	0.0135	Inf	0.592	0.644

Contrasts



› `marginalEffects avg_predictions()`

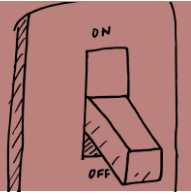
› code

```
model_predictions <- avg_predictions(  
  glmer_model,  
  by = 'condition'  
)
```

› output

condition	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
fear	0.393	0.0151	26.1	<0.001	496.2	0.364	0.423
neutral	0.595	0.0108	55.0	<0.001	Inf	0.574	0.616

Contrasts



- › Computing contrasts, supply additional arguments

› emmeans

```
emmeans(  
  glmer_model,  
  pairwise ~ age * trial *  
  condition,  
  regrid = 'response'  
)
```

› output

› marginaleffects

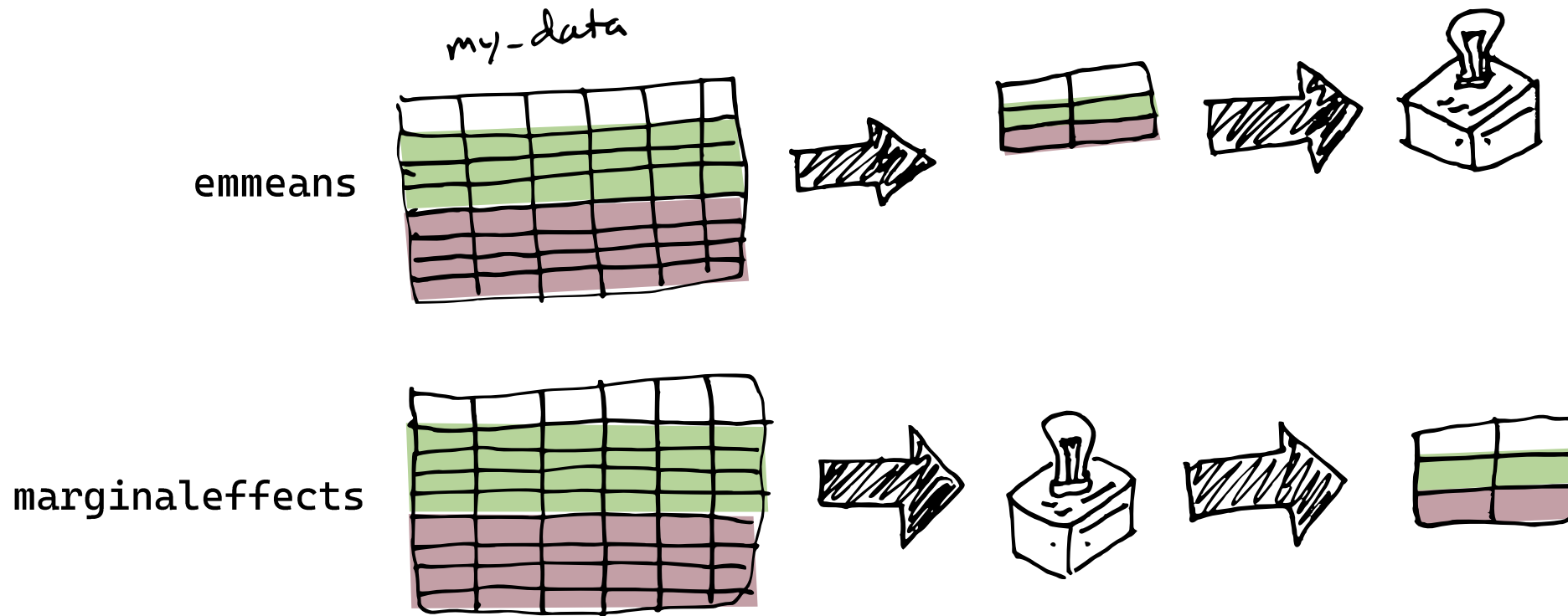
```
avg_predictions(  
  glmer_model,  
  by = 'condition',  
  hypothesis = ~ pairwise  
)
```

› output

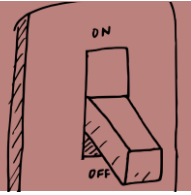
Hypothesis	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
(neutral) - (fear)	0.202	0.0165	12.2	<0.001	111.7	0.17	0.234

Why is this important?

- › These functions take a different approach to generating contrasts and predictions from a model



Contrasts

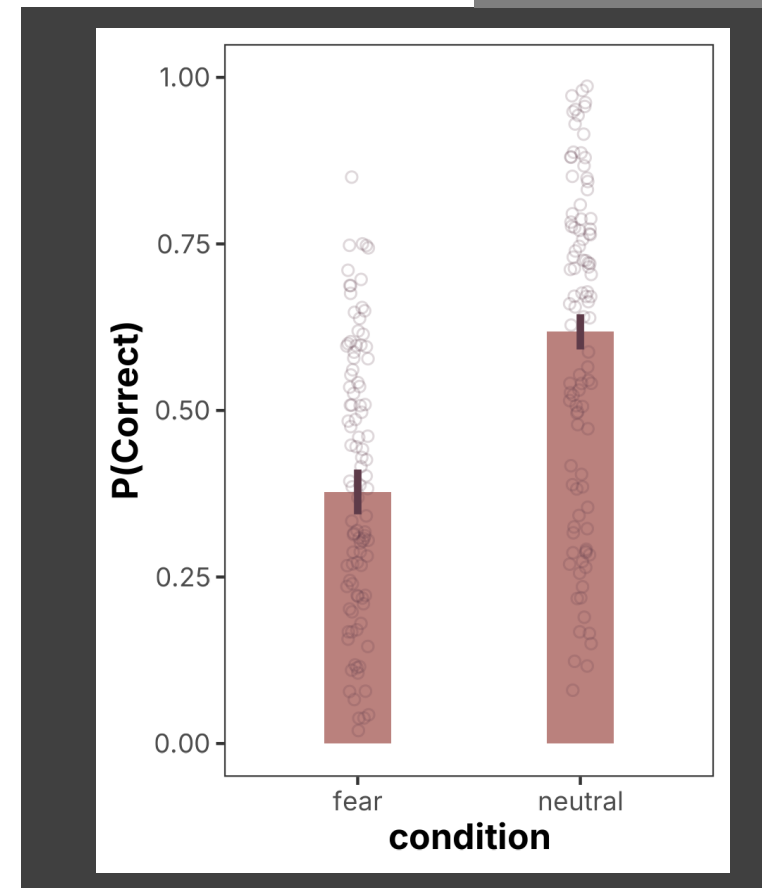


> visualization

> code

```
model_predictions ▷  
  # as_tibble() ▷ # if using emmeans  
  ggplot(aes(x = condition, y = estimate)) +  
  geom_col() +  
  theme_stuff()
```

> output



Marginal effects



› emmeans emtrends()

› code

```
emtrends(  
  glmer_model, ~ x_age, var = 'x_age',  
  at = list(x_age = c(-1, 0, 1)),  
  regrid = 'response'  
)
```

› output

NOTE: Results may be misleading due to involvement in interactions

x_age	x_age.trend	SE	df	asympt.LCL	asympt.UCL
-1	-0.158	0.00962	Inf	-0.177	-0.139
0	-0.199	0.01230	Inf	-0.223	-0.175
1	-0.180	0.00817	Inf	-0.196	-0.164

Marginal effects



› `marginalEffects avg_slopes()`

› code

```
avg_slopes(  
  glmer_model,  
  variables = 'x_age',  
  newdata = datagrid(x_age = c(-1, 0, 1), condition = unique)  
)
```

› output

x_age	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
-1	-0.149	0.0143	-10.39	<0.001	81.6	-0.177	-0.121
0	-0.163	0.0186	-8.79	<0.001	59.2	-0.199	-0.127
1	-0.145	0.0131	-11.04	<0.001	91.8	-0.171	-0.119

Marginal effects

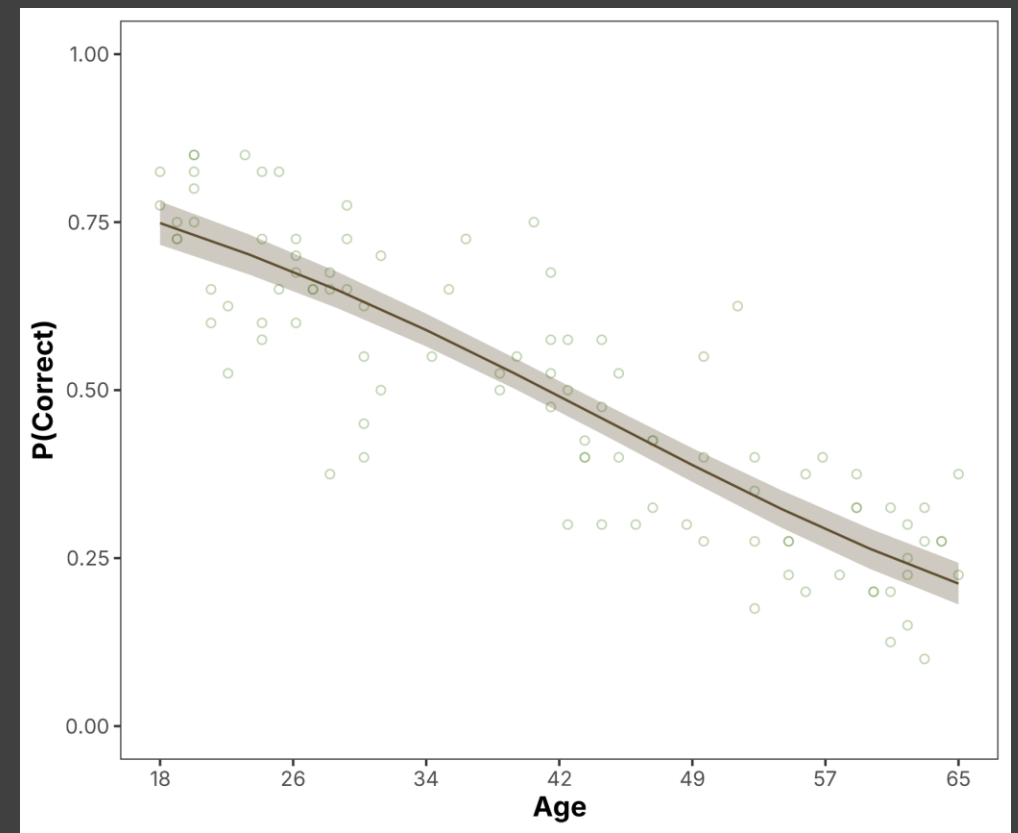


- › Visualization, use predictions

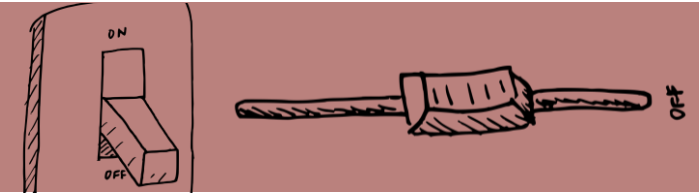
› code

```
preds >
  ggplot(
    aes(x = x_age, y = estimate)
  ) +
  geom_line() +
  geom_ribbon(
    aes(
      ymin = conf.low,
      ymax = conf.high
    )
  ) +
  theme_stuff()
```

› output



Interactions



- › Log odds slopes by condition across age

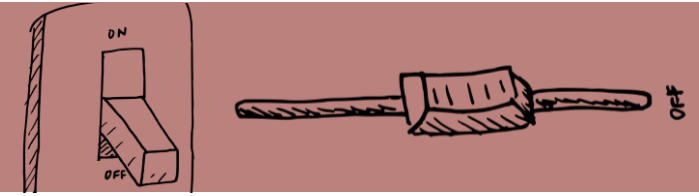
› emmeans

› marginaeffects

› output

condition	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
fear	-0.653	0.0743	-8.78	<0.001	59.1	-0.798	-0.507
neutral	-1.039	0.0612	-16.99	<0.001	212.6	-1.159	-0.919

Interactions

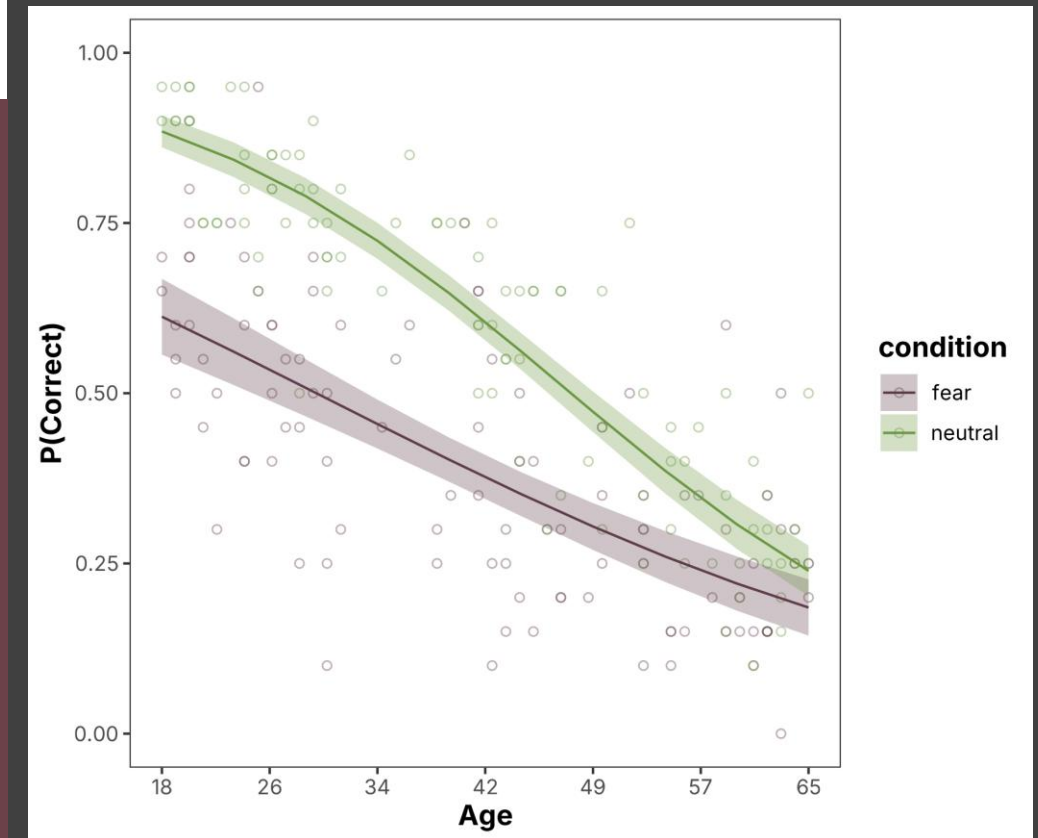


› Visualization, use predictions

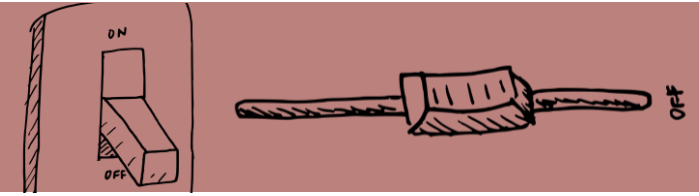
› code

```
preds >
  ggplot(
    aes(x = x_age, y = estimate)
  ) +
  geom_line(aes(color = condition)) +
  geom_ribbon(
    aes(
      fill = condition,
      ymin = conf.low,
      ymax = conf.high
    )
  ) +
  theme_stuff()
```

› output



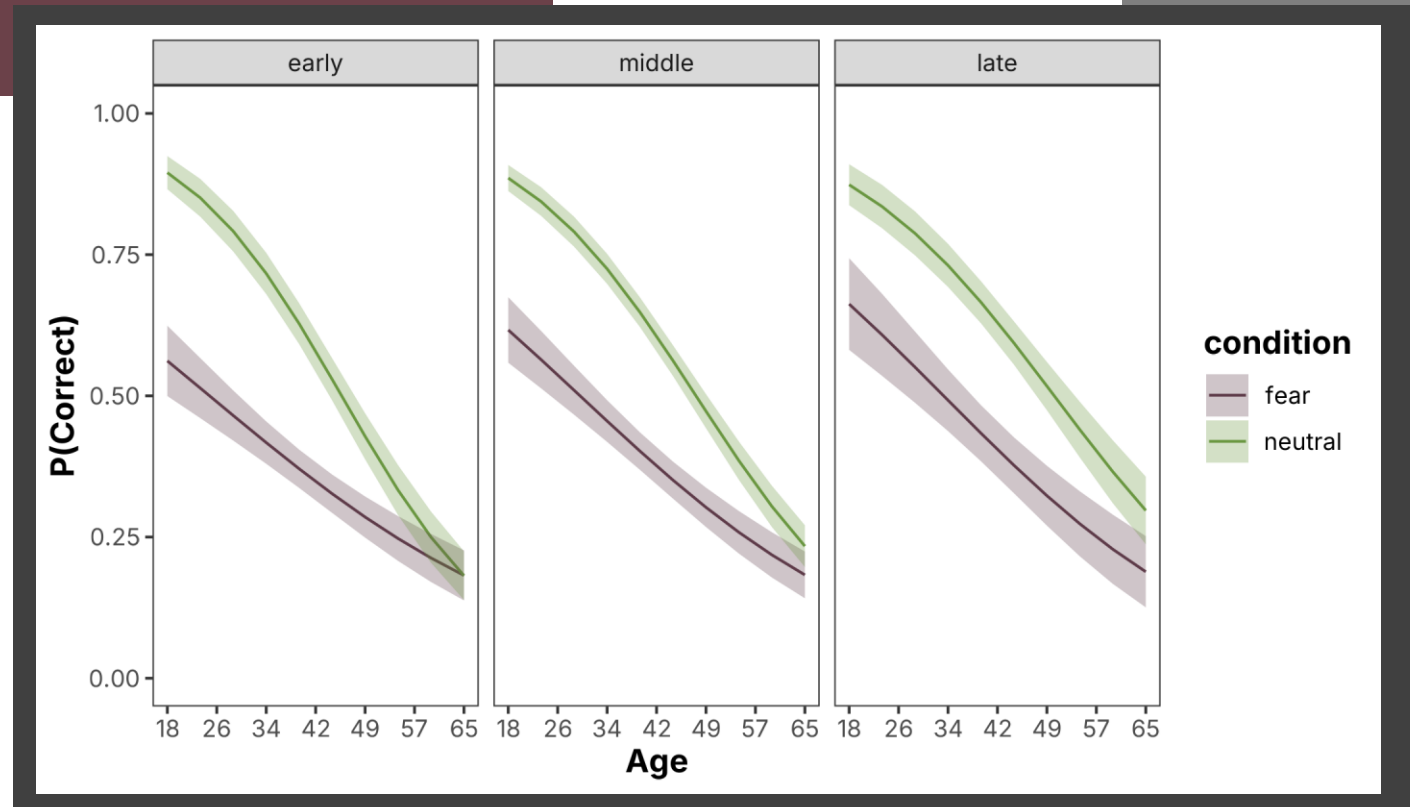
Interactions



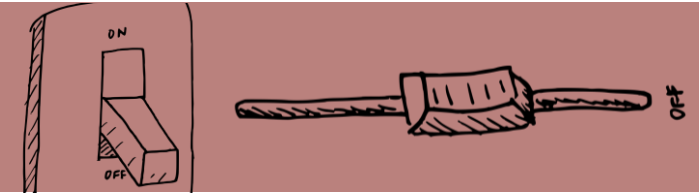
```
preds >
  ggplot(aes(x = x_age, y = estimate)) +
  geom_line(aes(color = condition)) +
  facet_wrap(~ x_trial) +
  theme_stuff()
```

> output

> code



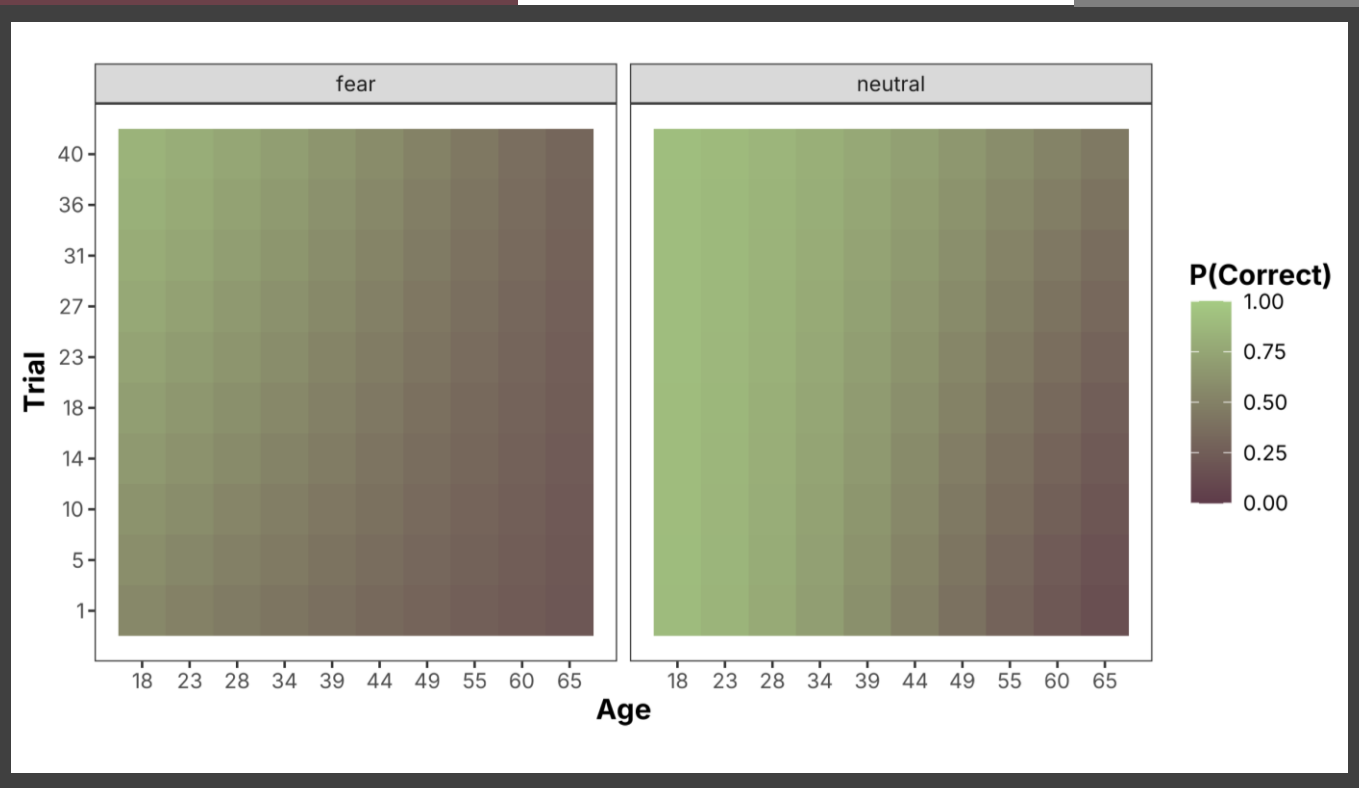
Interactions



```
preds ▷  
  ggplot(aes(x = x_age, y = x_trial)) +  
  geom_tile(aes(fill = estimate)) +  
  facet_wrap(~ condition) +  
  theme_stuff()
```

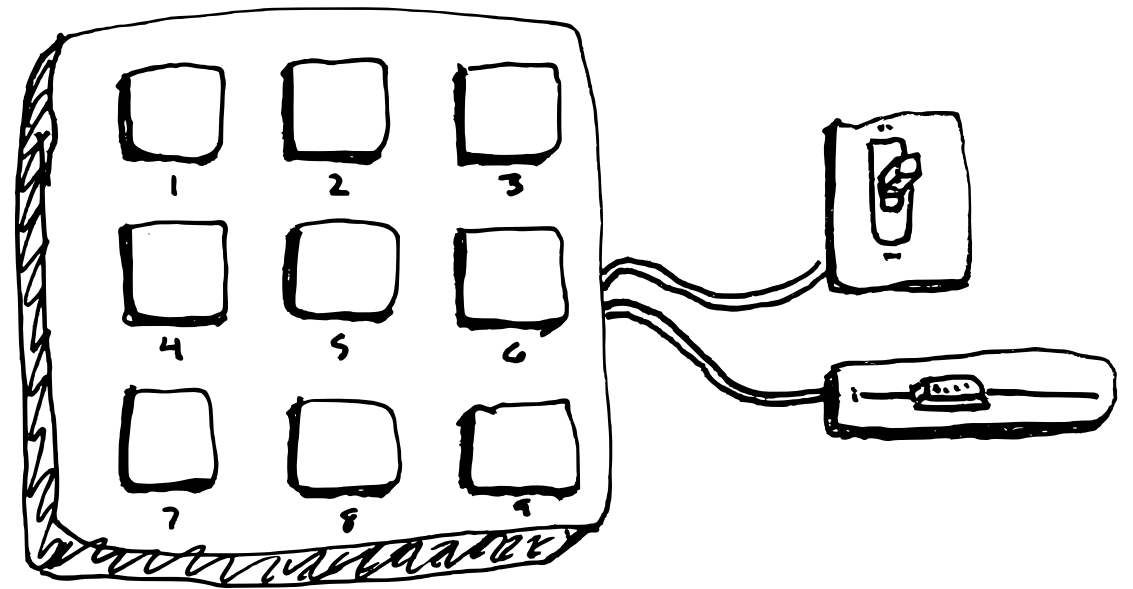
> code

> output

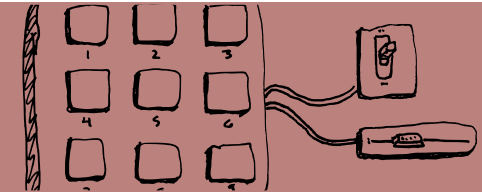


Extras

- › These functions can also produce marginal effects and contrasts for Bayesian models, carrying forward posterior uncertainty
- › Also do extra random effects handling either setting to zero or (EMM) integrating them out (ME)



Bayesian

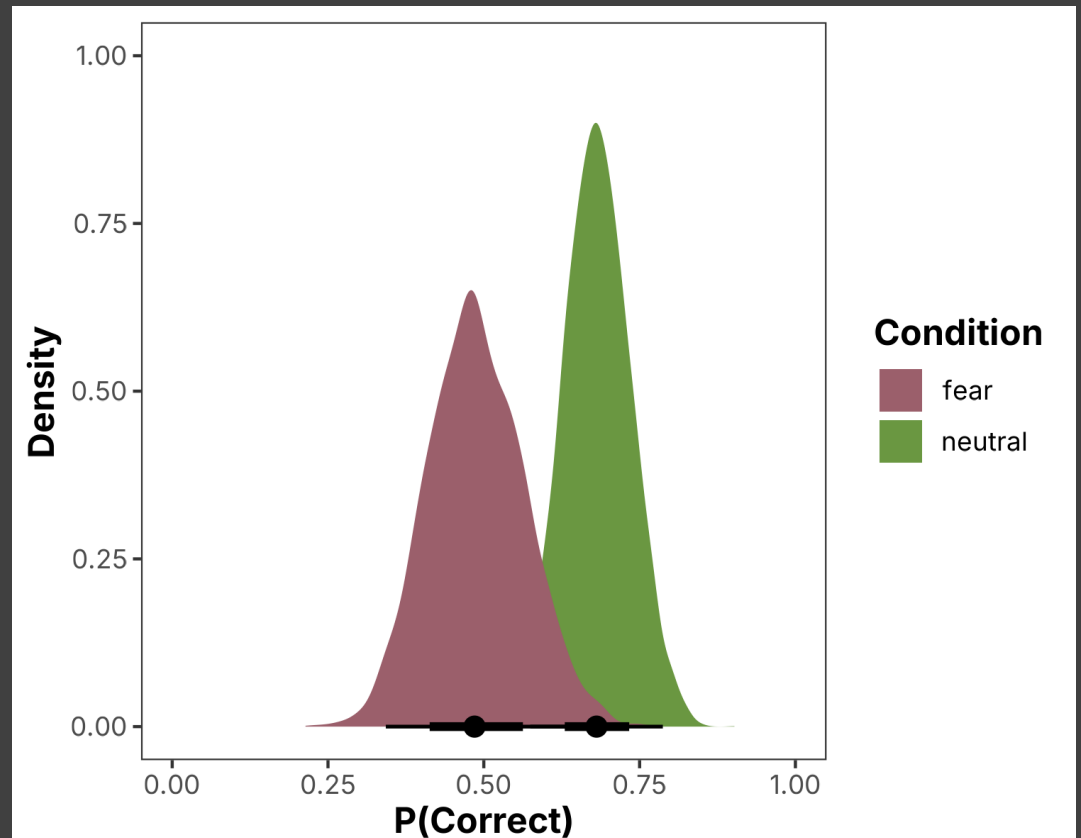


- › Contrast treatment conditions but as Bayesian posteriors

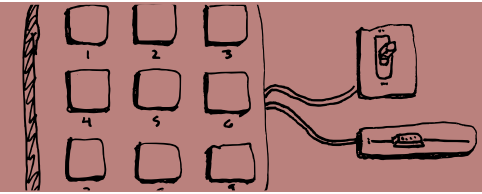
> output

> code

```
preds ▷  
  ggplot(  
    aes(x = draw, fill = condition)  
  ) +  
  stat_halfeye() +  
  theme_stuff()
```



Bayesian

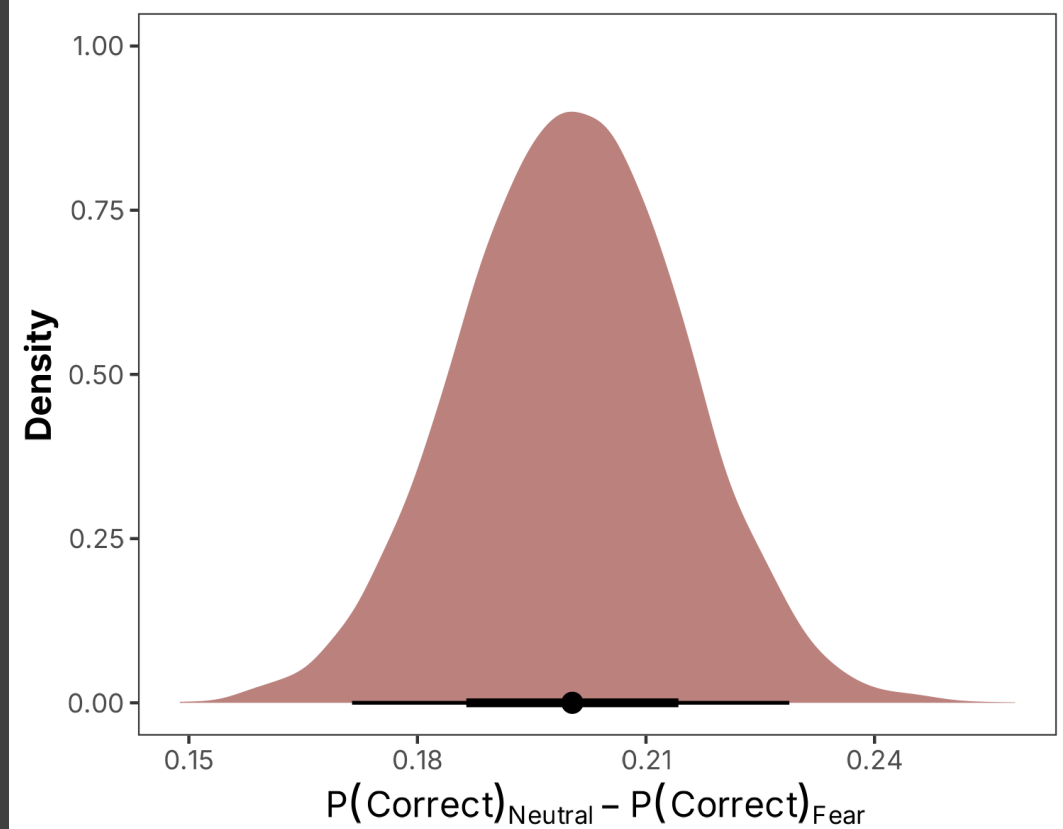


- › Get the treatment effect as its own distribution

> output

> code

```
preds ▷  
  ggplot(  
    aes(x = draw, fill = condition)  
  ) +  
  stat_halfeye() +  
  theme_stuff()
```



Takeaways

- › Work with your model like you're curious about what it does
 - › Flip its switches and move its sliders one at a time and visualize what it comes up with
- › We can accomplish this with models we fit in R with **marginalEffects** and **emmeans**
- › Additional info and source code can be found in the github link Morgan sent along in the announcement email!

Thank you!

